

សាកលវិទ្យាល័យភូមិន្ទភ្នំពេញ

ROYAL UNIVERSITY OF PHNOM PENH

ប្រធានបទរបាយការណ៍ស្រាវជ្រាវ

Loan Default Prediction

A Research Report
In Partial Fulfilment of the Requirement for the Degree of
Bachelor of Engineering in Data Science and Engineering

Tham Veasna

June 2025

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ABSTRACT IN KHMER

ការព្យាករណ៍ការបរាជ័យក្នុងការបង់ប្រាក់កម្ចីគឺជា ផ្នែកមួយយ៉ាងសំខាន់នៅក្នុងវិស័យហិរញ្ញវត្ថុព្រោះវាជួយក្នុងការបន្ថយហានិភ័យ និងរក្សាស្ថិរភាព។ នៅក្នុងប្រទេសកម្ពុជា ការប្រើប្រាស់នូវមធ្យោបាយចាស់ៗដើម្បីព្យាករហានិភ័យក្នុងការបង់ប្រាក់កម្ចីនេះតែងមានការលំបាកក្នុងការគ្រប់គ្រង និងឆែកព័ត៌មានច្រើនពីអ្នកខ្ចី។ ដូច្នេះហើយកិច្ចការស្រាវជ្រាវមួយនេះនិយាយពីការប្រើប្រាស់បច្ចេកទេសមួយចំនួន ដើម្បីពន្លឿនការព្យាករណ៍ការបរាជ័យក្នុងការបង់ប្រាក់កម្ចី និងការអនុម័តប្រាកកម្ចីអោយដំណើរការលឿន, រលូន, និងមានការជឿទុកចិត្តខ្លាំងជាងមុនសម្រាប់ធនាគារក្នុងប្រទេសកម្ពុជា។

កិច្ចការស្រាវជ្រាវនេះបានប្រើទិន្នន័យរបស់ Lending Club, ដែលមានចំនួនប្រហែល២,៧០០,០០០ ទិន្នន័យ ដែលបានឆ្លងកាត់ការសម្អាតនិងជ្រើសរើសទិន្នន័យដែលចាំបាច់។ ទិន្នន័យនៃហេតុការណ៍មួយចំនួនដូចជាវិបត្តិខាងហិរញ្ញវត្ថុពិភពលោកនៅឆ្នាំ២០០៨ និងការកើតឡើងនូវវីរុសកូវីដ-១៩ ត្រូវបានដកចេញដើម្បីបន្ថយភាពលំអៀង ហើយទិន្នន័យមួយចំនួនត្រូវបានកែប្រែដើម្បីអោយសមស្របជាមួយម៉ូដែល។ ម៉ូដែលដែលបានប្រើប្រាស់ក្នុងការស្រាវជ្រាវនេះមាន Logistic Regression, LightGBM, និង Neural Network និងការវាយតម្លៃតាមរង្វាស់ដូចជា Accuracy, AUC Score និង Confusion Matrix ដើម្បីផ្ទៀងផ្ទាត់ការសមត្ថភាពព្យាកររបស់ម៉ូដែល។

លទ្ធផលនៃម៉ូដែល LightGBM ដំណើរការបានល្អជាងម៉ូដែលដទៃទៀត ដោយទទួលបាន Accuracy ចំនួន 98.34% និង AUC ចំនួន 0.995 ក្នុងការស្រាវជ្រាវទីពាទាក់ទងជាមួយការប្រើប្រាស់ទិន្នន័យស្រដៀងទៅនឹងទិន្នន័យរបស់ធនាគារនៅប្រទេសកម្ពុជា។ ចំណែកឯការស្រាវជ្រាវទី១វិញ បានបង្ហាញពីបញ្ហាមួយចំនួនដូចជា overfitting និងdata leakage ដែលត្រូវអោយមានការប្រយ័ត្នក្នុងការរៀបចំនិងផ្ទៀងផ្ទាត់ទិន្នន័យ។ ការស្រាវជ្រាវនេះក៏បានបង្ហាញពីកម្រិតនៃការពឹងផ្អែកតែនៅលើ structured data ដែរ ដែលធ្វើអោយមានន្ទវតម្រូវការនៃការរួមបញ្ចូលនៃទិន្នន័យប្រភេទផ្សេងគ្នា ដូចជា unstructured textual data ជាដើម ដែលអាចបង្កើនសមត្ថភាពរបស់ម៉ូដែលអោយបានល្អជាងនឹងទៅទៀត។

កិច្ចការស្រាវជ្រាវនេះ ចូលរួមចំណែកជួយដល់ការអភិវឌ្ឍការវិភាគនៅក្នុងវិស័យហិរញ្ញវត្ថុដោយបានបង្ហាញពីការប្រើប្រាស់បច្ចេកទេសទាំងនេះទៅដោះស្រាយបញ្ហាសំខាន់ៗក្នុងការព្យាករណ៍ការបរាជ័យក្នុងការបង់ប្រាក់កម្ចី។ ម៉ូដែលដែលបានលើកឡើងក្នុងការស្រាវជ្រាវនេះ មិនត្រឹមតែអាចសម្រួលដល់ការការវាយតម្លៃទៅលើកម្ចីនោះទេ វាក៏អាចជួយទ្រទ្រង់ការផ្តល់នូវប្រាក់កម្ចីអោយមានជំនឿចិត្តនិងស្ថិរភាពក្នុងប្រព័ន្ធហិរញ្ញវត្ថុ។

ABSTRACT IN ENGLISH

Loan default prediction is a critical area for financial industry, helping them reduce risks and maintain stability. In growing markets like Cambodia, traditional methods for assessing loan risks often struggle to handle large amounts of data or account for detailed borrower behavior. This study explores how machine learning and deep learning techniques can improve loan default prediction and make the loan approval process faster and more reliable for Cambodian banks.

The research employs the Lending Club dataset, encompassing over 2.7 million records, which was meticulously prepared through data cleaning, feature selection, and feature engineering. Historical events like the 2008 financial crisis and the COVID-19 pandemic were excluded to minimize biases, and categorical variables were transformed for model compatibility. Three models, Logistic Regression, LightGBM, and Neural Networks, were implemented and evaluated using metrics such as accuracy, AUC and confusion matrix, to assess their predictive capabilities.

Results indicate that neural network outperformed other models, achieving a weighted accuracy of 98.34% and 0.995 AUC in experiment 3 which is aligning model features with those commonly used by local financial institutions. However, experiment 1 highlighted challenges such as overfitting, underscoring the need for careful data preparation and validation. The study also reveals the limitations of relying solely on structured data, advocating for the integration of multimodal data, including unstructured textual information, to further enhance model robustness and generalizability.

This research contributes to the advancement of predictive analytics in the financial sector by demonstrating the practical application of machine learning and deep learning techniques to address critical challenges in loan default prediction. The proposed models not only streamline the loan evaluation process but also support sustainable lending practices, fostering trust and stability in the financial ecosystem.

SUPERVISOR'S RESEARCH SUPERVISION STATEMENT

TO WHOM IT MAY CONCERN

Name of program: Bachelor of Data Science and Engineering
Name of candidate: Tham Veasna

Title of research report: Loan Default Prediction

This is to certify that the research carried out for the above-titled bachelor's research report was completed by the above-named candidate under my direct supervision. The material in this research report has not been used for any other degree. I contributed to the preparation of this research report as follows:……...
.....
.....
.....

Supervisor's name: ……

Supervisor's signature: ……

Date……

CANDIDATE'S STATEMENT

TO WHOM IT MAY CONCERN

This is to certify that the research report I, Tham Veasna, hereby present, entitled "Loan Default Prediction" for the degree of Bachelor of Engineering at the Royal University of Phnom Penh, is entirely my own work. Furthermore, it has not been used to fulfill the requirements of any other qualification, in whole or in part, at this or any other university or equivalent institution.

For the degree of Bachelor of Engineering at the Royal University of Phnom Penh is entirely my own work and, furthermore, that it has not been used to fulfill the requirements of any other qualification in whole or in part, at this or any other University or equivalent institution.

No reference to, or quotation from, this document may be made without the written approval of the author.

Signed by Tham Veasna: ……

Date: …….

Sign by Supervisor: Ms. Chea Daly

Supervisor's signature: ……..…

Date……..……

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Best regards,
THAM VEASNA

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LIST OF ABBREVIATIONS

RFE	Recursive Feature Elimination
SMOTE	Synthetic Minority Oversampling Technique
DT	Decision Tree
SVM	Support Vector Machine
LR	Logistic Regression
NB	Naïve Bayes
P2P	Peer to Peer
AUC	Area Under Curve
LDA	Latent Dirichlet Allocation
AdaBoost	Adaptive Boosting
LightGBM	Light Gradient Boosting Machine
KNN	K-Nearest Neighbor

CHAPTER 1
INTRODUCTION

1.1 Background to the Study

Around the world, banks and financial institutions lend money to individuals and businesses to support their growth and meet financial needs. However, when loans go unpaid—referred to as loan defaults—it not only impacts the institutions that provided the loans but can also harm the broader economy. Loan defaults reduce the capital available for future lending, affect interest rates, and can increase financial instability. Therefore, predicting the likelihood of loan defaults is crucial for maintaining a healthy lending ecosystem.

In Cambodia, where personal and business loans are becoming increasingly common, effectively predicting loan defaults can benefit both banks and borrowers. For financial institutions, it enables them to assess the risk associated with each loan application, allowing for informed decision-making that reduces the chances of loss. For borrowers, accurate assessments can lead to fairer loan terms, as banks would feel more secure about repayment reliability. By improving prediction methods, Cambodian banks can enhance lending practices, ensuring that loans support economic growth while minimizing financial risk. This approach not only stabilizes the financial system but also fosters trust in the lending process, helping more people and businesses access the financial resources they need responsibly.

1.2 Problem Statement

When loans default, banks incur financial losses that can disrupt their normal operations and impact the availability of credit. These losses may force banks to limit lending, increase interest rates, or tighten lending criteria, which in turn can slow down economic growth. At a broader level, high default rates can ripple through the country's economy, affecting the stability of the financial sector and ultimately influencing the living standards of the population. A weakened financial sector reduces funding for businesses, decreases job creation, and can drive up costs for everyday borrowers, impacting household finances and general economic health.

This issue is especially pertinent as lending activity continues to rise in Cambodia. With increasing access to credit, there is a growing need for banks to understand and

mitigate the risks associated with loan defaults. Traditional methods for assessing default risk may no longer be sufficient in an evolving market, as they often rely on limited data or are unable to capture nuanced borrower profiles. Therefore, there is a critical demand for more accurate and robust predictive models that can identify potential defaults early on. Enhanced prediction capabilities will enable Cambodian banks to safeguard their assets while continuing to expand lending opportunities in a way that is both secure and sustainable. This research seeks to contribute to this effort by addressing the need for improved risk management tools in Cambodia's lending sector, ultimately supporting economic resilience and financial stability.

1.3 Aim and Objective

This study aims to develop a model designed to assist bank operators, particularly those responsible for loan approval, in streamlining the loan evaluation process. By leveraging borrower historical data and advanced machine learning techniques, the model will provide quick and accurate assessments of loan default risks. This approach will help reduce the time spent on manual evaluation, allowing operators to focus on higher-level decision-making tasks and improving operational efficiency.

The model will enable operators to make faster, data-driven decisions based on predictive analytics, minimizing human error and bias. It will analyze various factors in a borrower’s history, such as payment behavior, financial stability, and other key indicators, to assess the likelihood of loan default. By providing actionable insights, the model will not only speed up the loan approval process but also ensure that banks can make more informed and reliable decisions, improving the overall quality of lending operations.

Ultimately, the aim is to enhance the lending process by offering a scalable, efficient, and effective tool that empowers banks to manage loan risks while maintaining a smooth, efficient workflow for loan officers and operators.

1.4 Rationale of the Study

The process of determining whether a person is likely to default, as used by Cambodia financial institutions, often relies on a scoring system such as the K-Score, which is similar to the FICO score developed by the Fair Isaac Corporation in America. This scoring system evaluates individuals based on a range from low to high scores. However, loan approvals still require a human decision-maker at the institution to assess and finalize the decision. This study aims to assist financial institutions by automating the identification

of potential defaulters among a large number of applicants, thereby facilitating quicker and more informed decision-making.

1.5 Limitation and Scope

This study primarily focuses on loan approval operations for banks within Cambodia, aiming to develop a predictive model that can assist in evaluating loan default risks. However, one of the key limitations of this research is the limited access to real, representative data from financial institutions in the country. This lack of access may restrict the model's ability to capture the full spectrum of financial behaviors and loan characteristics specific to the Cambodian market.

Without authentic borrower data reflecting the local demographic trends, economic conditions, and unique lending practices, the model's performance may not accurately mirror the realities faced by banks operating in Cambodia. The absence of such comprehensive data can hinder the model's ability to identify patterns and risk factors that are highly relevant to the Cambodian context, potentially leading to suboptimal decision-making.

Additionally, financial data can vary significantly between institutions due to differences in lending policies, borrower profiles, and regional economic factors. As a result, the model may struggle to generalize effectively across different banks or even fail to capture local nuances, limiting its broader applicability. To address this limitation, access to more diverse, high-quality data from local financial institutions would be essential for improving the model's precision and enhancing its practical utility in real-world loan approval scenarios.

1.6 Structure of Study

This study is organized into six chapters, each building on the previous to provide a comprehensive understanding of the research.

The first chapter introduces the research by presenting the background and context of the study. It also articulates the problem statement that needs to address, defines the objectives to be achieved, and outlines the limitations and scope. By providing these foundational elements, this chapter sets the stage for the subsequent discussions and analyses.

The second chapter focuses on the literature review, offering an overview of the research topic and its significance. It examines relevant prior studies to identify the current state of knowledge and practices in the field. Furthermore, this chapter highlights research gaps and opportunities that motivate and justify the need for this study.

The third chapter explains the methodologies and tools employed in this research, detailing the processes of data collection, preparation, and modeling. Data preparation involves cleaning, feature selection, and engineering to ensure high-quality inputs for the models. The methodologies are chosen after many experiments and exploring to address the problem statement, ensuring that the study achieves its objectives. This chapter provides a clear and structured framework that guides the experimentation phase.

The fourth chapter is dedicated to the experimentation and results process, describing the implementation and evaluation of various models used in this study. It details the design and execution of experiments, ensuring reproducibility and reliability of results. The chapter also explores the technical aspects of the experimentation, such as model training, hyperparameter tuning, and configuring different cleaning algorithms to maximize the performance of the models.

The fifth chapter presents the results and engages in a detailed discussion of the findings. It showcases the outcomes of the training process. Additionally, this chapter interprets the findings, offering insights into their practical implications and addressing any unexpected results or challenges encountered.

Finally, the sixth chapter concludes the study by summarizing the key findings and their contributions to the field. It acknowledges the limitations of the research and discusses how these may have influenced the results. Furthermore, this chapter identifies potential areas for future work, suggesting ways to build on the current research to address remaining challenges and explore new opportunities in the field.

CHAPTER 2
LITURATURE REVIEW

2.1 Overview of Loan Default Prediction

Loan default prediction has been a critical area of research, with various methods explored to enhance model performance, accuracy and decision-making efficiency for financial institutions. This section reviews three past studies that have contributed to this field, with machine learning, and deep learning approaches.

2.2 Previous Studies

2.2.1 Overview of Existing Applications

Existing research has demonstrated various machine learning approaches for loan default prediction. Multiple studies have explored different datasets and algorithms to improve prediction performance. The following studies highlight contributions that researchers have made in this field:

2.2.2 Zhima Credit Score in Loan Default Prediction for Personal Loans

This study, conducted by Haomin Wang et al. (2021), was carried out by a research team from University of Electronic Science and Technology in Chengdu, China. The study focuses on applying Zhima Credit Score to models trained on a Chinese online loan platform dataset, which consists of 28,698 data points with 8 attributes.

The researchers used SMOTE to balance dataset by expanding the minority class to an optimal ratio. The dataset is then trained on models such as C4.5, RF, NB, KNN, SVM, NN. The models are evaluated by 10 fold cross validation using Weka software involving metrics such as Accuracy, AUC, Precision, Recall, Kappa Statistic, Mean Absolute Error (MAE).

The result of these experiments are:

- C4.5 92.64% accuracy, 0.942 AUC
- RF 94.55% accuracy, 0.970 AUC
- NB 74.12% accuracy, 0.813 AUC
- KNN 88.85% accuracy, 0.924 AUC
- SVM 84.38% accuracy, 0.834 AUC
- NN 81.75% accuracy, 0.892 AUC

2.2.3 Combining Soft Information Extracted from Descriptive Text

This study, published by Jiang et al. (2018), used an LDA model to extract both hard and soft information, which was then fed into machine learning models including LR, NB, SVM, and RF.

The researchers worked with a dataset from Eloan, a major P2P lending platform in China, containing 132,000 records with 27 features from January 2014 to January 2016, of which 9 features were personal information, 11 were financial status, and 6 were loan information.

They used the LDA model to extract topics from text that accompanied loan applications. They calculated the probability distribution of the text in each topic to find the number of occurrences, determining that the model worked best with 15 topics, of which 6 had obvious meanings such as assets, income and expenses, work, family, business, and agriculture.

Then they used these features to train the four models mentioned above and evaluated them via five-fold cross-validation, keeping the splitting of folds identical across models.

The results showed the following performance:

- LR: 85.2% accuracy, 0.797 AUC
- NB: 77.6% accuracy, 0.759 AUC
- SVM: 85.4% accuracy, 0.8 AUC
- RF: 86% accuracy, 0.84 AUC

2.2.4 New Aspect using Meta-level Phone Usage Data

This study, published by Lin Ma et al. (2018), was conducted by researchers from Xi'an, China who proposed using meta-level phone usage data as an alternative to traditional credit scoring methods. The study employed an AdaBoost model alongside LR and RF for performance comparison.

They found three distinct phone usage patterns - telecommunication patterns, mobility patterns, and app usage patterns - that were significantly related to loan default. These patterns were extracted from raw phone usage data of borrowers combined with their actual loan records.

The researchers utilized a dataset of 3,027 records and used the RFE method to select the optimal variable combination from the raw data. However, they found that a bias existed in the feature elimination process due to randomly selecting features and samples. Therefore, they applied a synthetic method to select features from multiple RFE results.

They then split the dataset into an 80:20 ratio for training and testing, and performed parameter tuning for AdaBoost with a learning rate of 0.1 and forest size of 500. Results showed:

- AdaBoost 71% accuracy, 0.723 AUC
- LR 67% accuracy, 0. 684 AUC
- RF 68% accuracy, 0.706 AUC

2.3 Research Gaps and Opportunities

2.3.1 Limitations in Existing Studies

The reviewed studies primarily focused on machine learning models (RF, DT, XGBoost, AdaBoost, NB, and SVM), with no exploration of deep learning techniques that could potentially enhance prediction performance.

Additionally, these studies relied on relatively small datasets for model training. The largest dataset mentioned contained only 132,000 records, which may be sufficient for classical machine learning models but could prove inadequate when applied to larger-scale data scenarios.

2.3.2 Opportunities for Research

First, expanding the dataset to incorporate millions of loan records from diverse financial institutions would significantly enhance the precision of default prediction models. Many existing studies, including this one, rely on limited sample sizes that may not fully capture the spectrum of borrower behaviors under different economic conditions. A larger dataset would strengthen model training, improve generalization to unseen cases, and allow for more accurate detection of uncommon yet critical patterns—such as defaults among typically low-risk borrowers.

Second, further investigation into advanced deep learning methods could lead to major gains in prediction accuracy, especially as datasets become larger and more complex. Although this study highlights the efficacy of simple neural networks, more sophisticated architectures—such as transformers and graph neural networks—have yet to be thoroughly

explored for loan default prediction. These techniques could more effectively capture intricate relationships in financial data, including connections between borrowers or nuanced trends in extended payment histories.

CHAPTER 3
METHODOLOGY AND TOOLS

3.1 Research Design

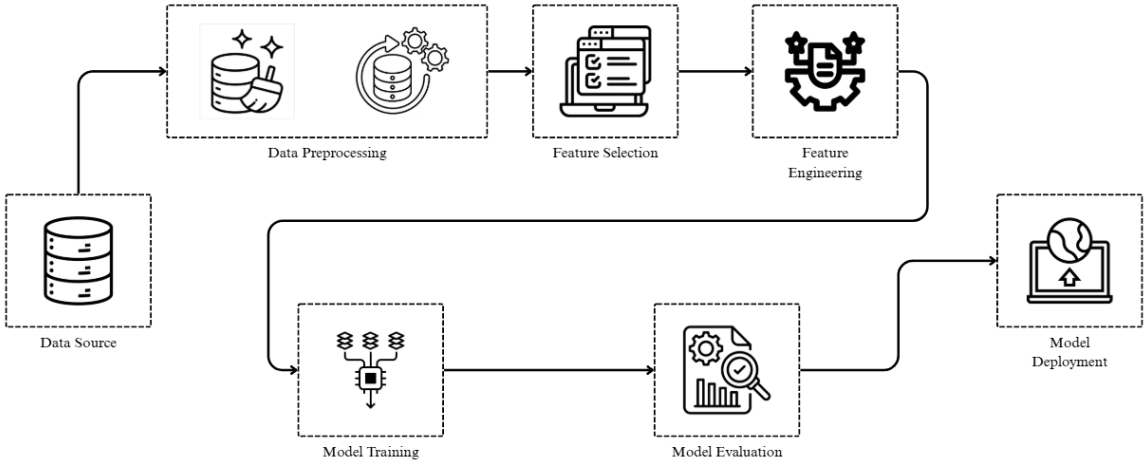


Figure 1: Research WorkFlow

This research follows the steps from data source to model deployment. At the start, we find a data source then we perform data preprocessing on the raw data and then select feature according to each experiment and then do feature engineering. Afterward the data is fit into our models and evaluate after training. Finally, we deploy the model on to a website.

3.2 Tools and Technologies Used

We used several tools that helped with this research study, like Visual Studio Code, Python, for coding and some main libraries like scikit-learn, TensorFlow, and Keras for models training and evaluation.

3.2.1 Visual Studio Code

Visual Studio Code was used as the programming environment for data preparation and running experiments.

3.2.2 Python

Python was used as a programming language for its compatibility with many machine learning libraries and interactive python notebook (.ipynb) file format.

3.2.3 Scikit-Learn

Scikit-Learn is a machine learning library for tasks like classification, regression, clustering and dimensionality reduction. It has many functions for data preprocessing, model evaluation, and contains supervised models like LR, DT, RF, SVM, and GB, etc.

3.2.4 TensorFlow

TensorFlow is a lightweight library that provides tools to easily build, train and deploy deep learning models across different platforms.

3.2.5 Keras

Keras is a high-level deep learning API that simplifies the process of building deep neural networks which makes it easier to define, train and evaluate deep learning models without requiring much knowledge of operations.

3.3 Data Source

For this study, we used the Lending Club dataset, which was scraped by Kaggle community members over the period from 2007 to September 2020 and is publicly available on Kaggle. This dataset consists of over 2.9 million records with 141 total features. Link to Dataset: https://www.kaggle.com/datasets/ethon0426/lending-club-20072020q1

3.4 Data Preparation

3.4.1 Data Cleaning

As we know, the world had two global crisis happened which connect to financial industry. Those are the global financial crisis and Covid-19, therefore it had to be removed. Most features with missing values are dropped and some are filled with data imputation.

3.4.2 Feature Selection

We used different features for each experiment using different methods. For experiment 1, we used feature importance to determine the features with more than 20% importance score.

For the second experiment, We chose features from one of the research paper we read with less data than ours and with the similar features therefore having 10 features and 1 label.

For the third experiment, We chose features based on CBC personal credit application form and add credit score, so we have a total of 21 features and 1 label.

3.5 Feature Engineering

In this section is about how we dealt with the data so it is suitable for model training. We took the selected features of each experiment to perform label encoding on categorical data and convert time series data to a suitable type for models and reduce memory usage for all features as much as possible.

3.6 Model Selection

In this research study, we used three different models to experiment with. These include LR, LightGBM, and neural network.

We first chose LR for its easy to use for training as it does not require the need to have a validation data for its training and that it does not have the need to have parameters.

We chose LightGBM because it is one of the machine learning models that is recommended for projects such as loan default, therefore we decided to use this model.

We chose neural network because we wanted to explore deep learning for this project and see what kind of performance and how good it is to perform on this project.

3.6.1 Logistic Regression

Logistic Regression is a statistical method used for binary classification to predict the probability of an occurring event. It categorizes outcomes based on one or more features into one of two classes, such as True/False or 0/1.

3.6.2 LightGBM

LightGBM is a machine learning model that uses gradient boosting method which constructs a strong learner by sequentially adding weak learners in a gradient descent manner.

3.6.3 Neural Network

Neural Networks are machine learning models that mimic the functions of human brain which consist of interconnected neurons that process data, learn patterns and assist with tasks such as recognition, detection, decision making, etc.

3.7 Model Evaluation Metrics

We used 3 metrics such as confusion matrix, accuracy and AUC to determine the performance of our models and for model comparison with exisiting applications.

3.7.1 Confusion Matrix

Confusion matrix is a tool used to evaluate the performance of a machine learning model, particularly for classification tasks. It provides summary of the model's predictions against actual outcomes, helping to identify the number of correct and incorrect predictions made by the model.

Confusion matrix have four components:

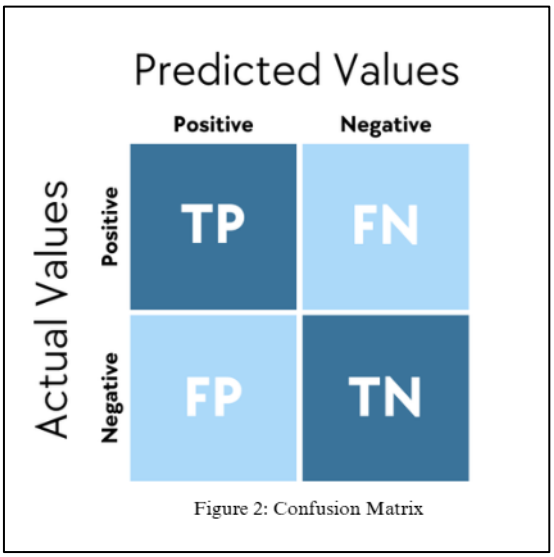


Figure 2: Confusion Matrix

- True Positive (TP): The model correctly predicted a positive instance (the actual outcome was also positive).
- True Negative (TN): The model correctly predicted a negative instance (the actual outcome was also negative).
- False Positive (FP): The model incorrectly predicted a positive outcome (the actual outcome was negative). This is also known as a Type I error.
- False Negative (FN): The model incorrectly predicted a negative outcome (the actual outcome was positive). This is also known as a Type II error.

3.7.2 Accuracy

Accuracy measures the proportion of correct predictions (both true positives and true negatives) out of the total number of predictions.

$$\text{Accuracy} = \frac{TP+TN}{TP+TN+FP+FN}$$

3.7.3 AUC

AUC is a graph used to evaluate the performance of a binary classification model. It measures how well the model separates positive cases from negative cases at different threshold levels.

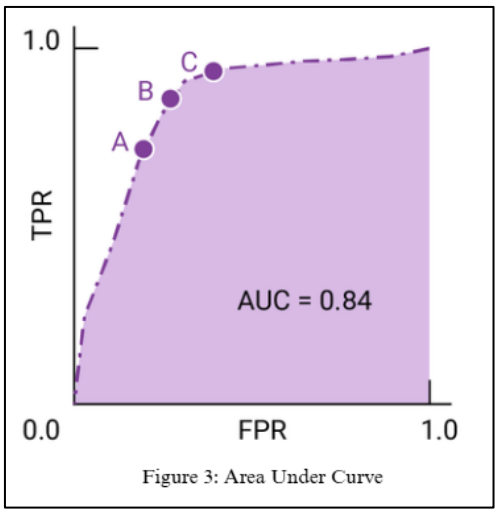


Figure 3: Area Under Curve

The graph itself have two classes:

- True Positive Rate (TPR): how often the model correctly predicts the positive cases (also known as Sensitivity or Recall).
- False Positive Rate (FPR): how often the model incorrectly predicts a negative case as positive.

The AUC score ranges from 0 to 1, where a score of 1 indicates excellent performance, a score of 0 means poor performance and a score of 0.5 means random guessing

CHAPTER 4
RESULTS

4.1 Experimental Setup

For our experiments, we use our own laptop to train and experiment with the models. Our laptop has Ryzen 7 6800H CPU, RAM of 32GB, Storage Capacity of 1TB and NVIDIA Geforce RTX 3060 as GPU.

4.2 Training Configuration

This part of the study is about what happened to the model during training of all experiments we had conducted. We initially used only LightGBM in all experiments except the third experiment.

For LR model, we set six parameters as the model is a relatively simple model compare to the other two models. We set a seed for randomness 'random_state' to 42, give the model 100 iterations, set smaller strict tolerance at 0.001, use 'solver' = 'sag' because it is fasterr for large data, set class weight to balance to avoid bias and have the model used all CPU cores.

For LightGBM, we set fourteen parameters into the model including learning_rate of 0.05. The other parameters are taken from the document of LightGBM and were experimented on different values until the results change less than before. Those parameters are objective, num_leaves, metric, min_child_weight, max_depth, min_child_samples, reg_alpha, reg_lambda, seed, n_estimators, subsample, colsample_bytree and subsample_freq.

As for Neural Network, we experimented on having different number of hidden layers and neurons and we used Adam optimizer with learning rate of 0.0001 and BinaryCrossentropy loss function.

4.3 Model Training and Performance

4.3.1 Experiment 1

In the first experiment, we performed feature importance to find the most contributed feature in the dataset. We used LightGBM classifier to train on a sample of 20% of the entire dataset and plot it out as in Figure 4. We removed features by setting a threshold of 40. The top 18 features excluding label were selected for model training.

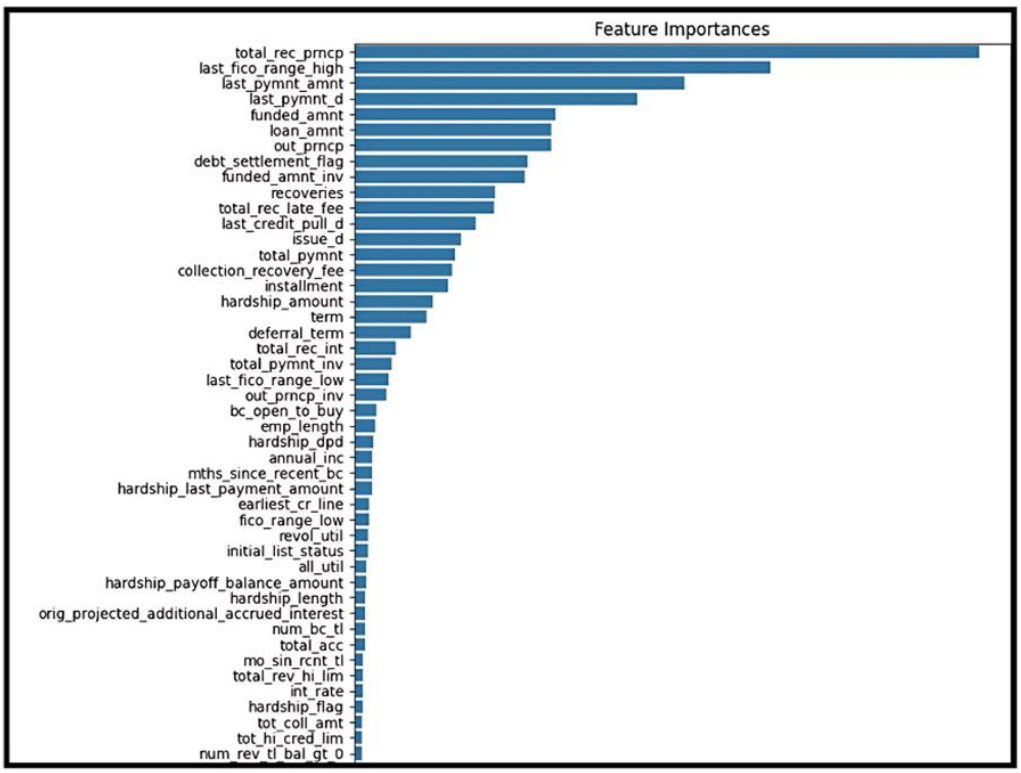


Figure 4: Feature Importances

Afterward, we trained the dataset with the selected features with the training configuration set on LightGBM. We then plot the learning curve of LightGBM. The model was trained for 15 minutes and 41 seconds.

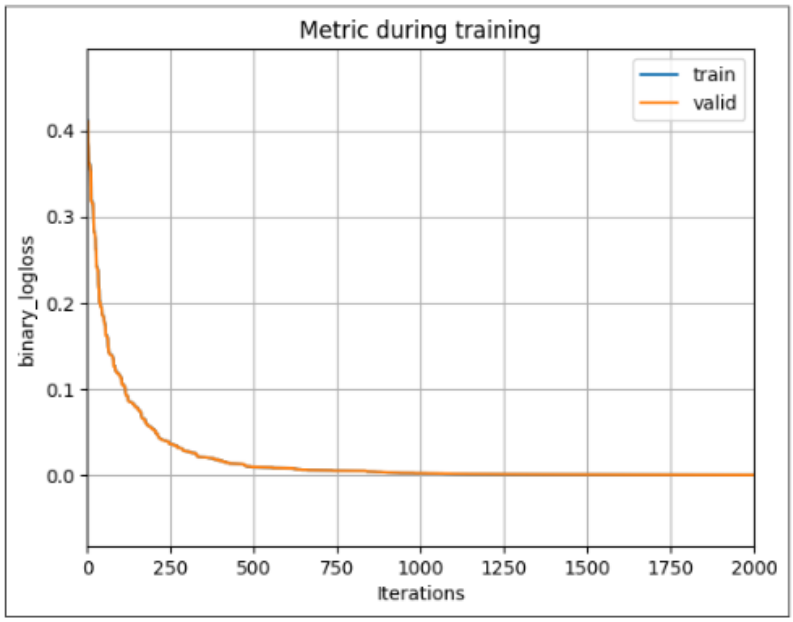


Figure 5: Experiment 1 LightGBM Learning Curve

The graph shows that the training and validation data used on these selected features has the exact same score for all 2000 iterations from the start with 40% loss to 0% loss.

```
Classification Report:
              precision    recall  f1-score   support

           0       1.00      1.00      1.00    181050
           1       1.00      1.00      1.00     30642

    accuracy                           1.00    211692
   macro avg       1.00      1.00      1.00    211692
weighted avg       1.00      1.00      1.00    211692

Accuracy: 1.00
```

Figure 6: Experiment 1 Classification Report

From Figure 6, we can see that the model performs really well which it got everything right for both classes and achieve 100% accuracy and perfect score on everything.

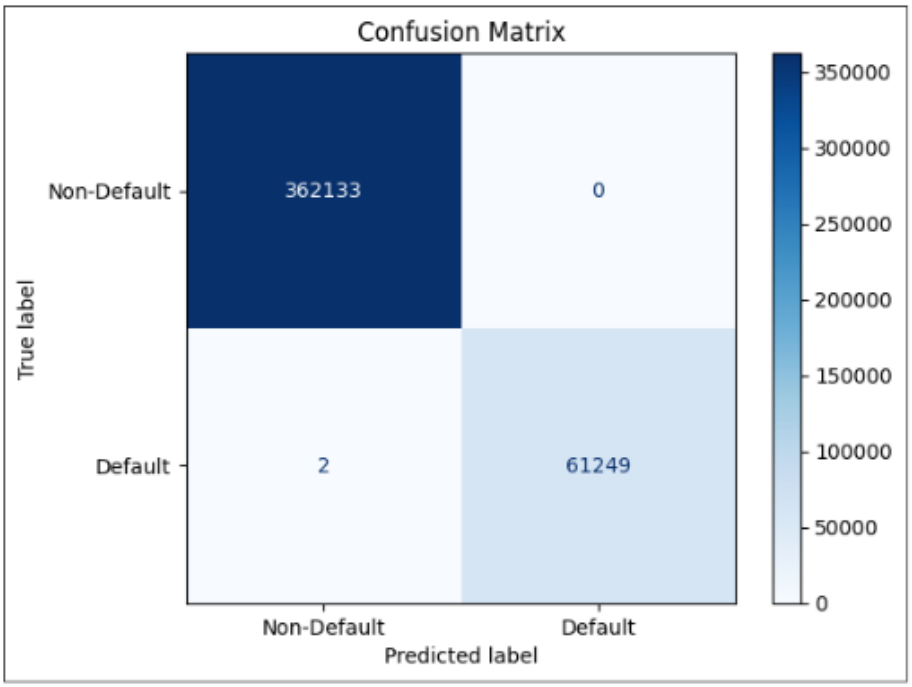


Figure 7: Experiment 1 Confusion Matrix

From Figure 7 below, we can see that the model did not guess all of them right so it should not be 100% accuracy. We suspected that the model is likely to overfit as the result from the classification report and confusion matrix does not make sense.

4.3.2 Experiment 2

In the second experiment, We used 10 features which is from a research study that we read as it used a smaller dataset than ours. We added stratified ten-fold cross validation to avoid bias on the majority class. The model was trained for 11 minutes and 59 seconds.

The research paper link:

https://www.sciencedirect.com/science/article/pii/S1877050919320277

Feature	Description
loan_amnt	Amount applied by borrower
Installment	Monthly payment owes by borrower
Grade	Loan grade assigned by Lending Club
Open_acc	Number of open credit lines in borrower credit files
Total_pymnt	Total payment of borrower
Total_rec_int	Interest on loans for borrower
Home_ownership	Status provided by the borrower during registration
Verification_status	Indicates if the co-borrower joint income was verified by Lending Club, not verified or if the source was verified
Application_type	Indicates whether the loan is individual application or joint
Purpose	Category provided by borrower for loan request
Term	Number of payments on loan. Values are in months and can be 36 or 60

Table 1: Experiment 2 Features

Using these features, we received the results as below:

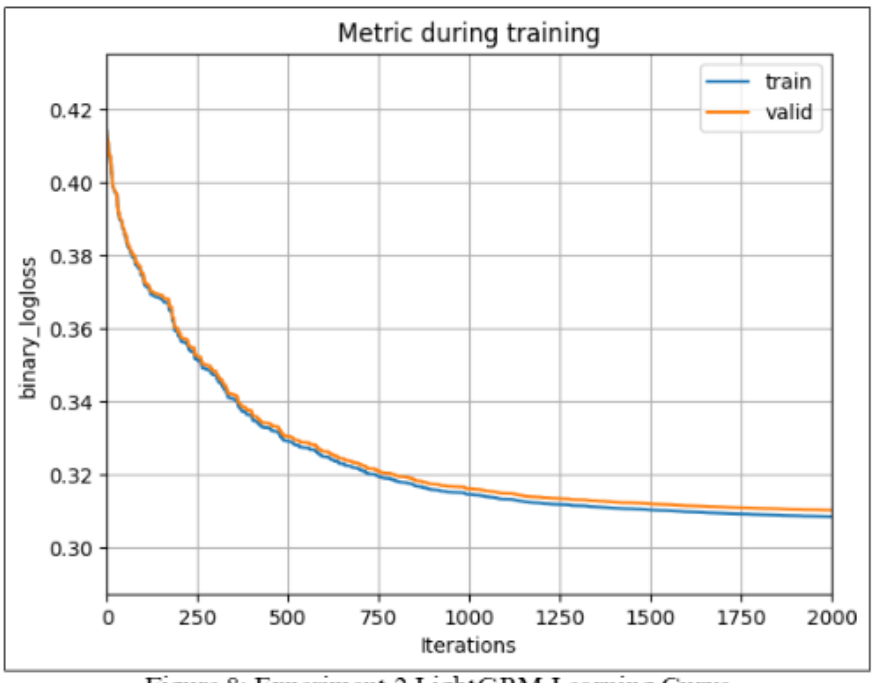


Figure 8: Experiment 2 LightGBM Learning Curve

This graph shows that the model performs quite well as it is almost the same value for both training and validation data, but it seems that after 2000 iterations the model still experience some a bit over 30% loss.

```
Classification Report:
              precision    recall  f1-score   support

           0       0.89      0.98      0.93    362133
           1       0.64      0.25      0.36     61251

    accuracy                           0.87    423384
   macro avg       0.76      0.61      0.64    423384
weighted avg       0.85      0.87      0.85    423384

Accuracy: 0.87
Test AUC: 0.6138866128573339
```

Figure 9: Experiment 2 Classification Report

From the classification report, we can see that the accuracy achieved for this model is 87% but the AUC score is so low as it is 0.6138. this can be seen from the F1-Score in the classification report that the non-default is mostly correct with 0.93 and the default is less than half are predicted correctly.

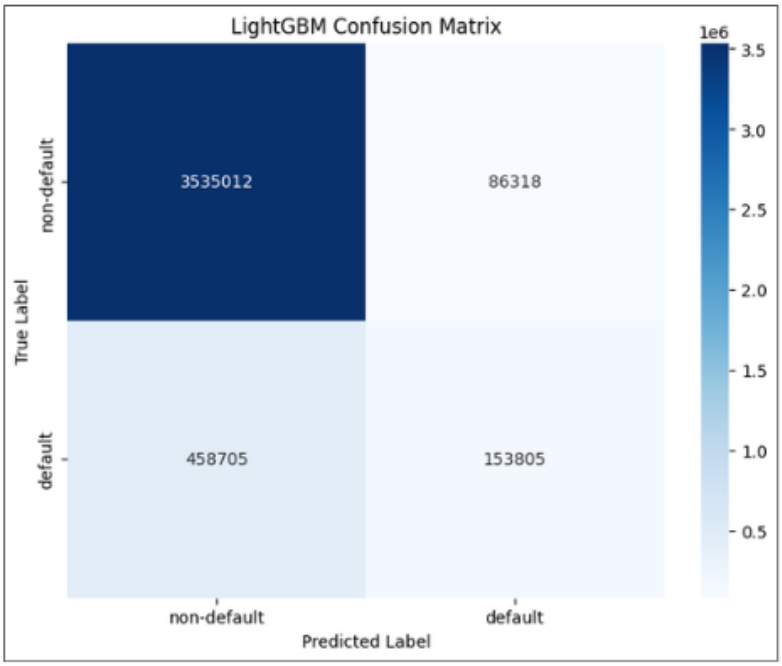


Figure 10: Experiment 2 Confusion Matrix

From the confustion matrix above, it seems that the model is performing quite well for non-default but not so much for default and the false negative and false positive are quite big in numbers.

4.3.3 Experiment 3

In the third experiment, we decided to add two other models alongside LightGBM to train on the features based on CBC personal credit application form to see if LightGBM is the best or is there a better alternative. The matched features are as follows:

Feature	Description
loan_amnt	Amount applied by borrower
Installment	Monthly payment owes by borrower
Funded_amnt	Amount that borrower get
Total_acc	Number of total credit lines in borrower credit files
Restructured_loan	Whether the loan has been restructured. Values are in No or Yes.
Int_rate	Interest rate of the loan
Annual_inc	Income of borrower in a year
Outstanding_balance	the principal amount borrowed and any accrued interest or fees
Application_type	Indicates whether the loan is individual application or joint
Purpose	Category provided by borrower for loan request
Term	Number of payments on loan. Values are in months and can be 36 or 60
Issue_date	The date when the loan is issued
Last_amnt_paid	The amount that the borrow paid last
Last_payment_date	The date of the payment that the borrow paid
Past_due_fee	The amount that the borrow pay for past the deadline
Next_payment_date	The date of the next payment that the borrow have to pay
Last_credit_pull_d	Date when credit was last pulled
Earliest_cr_line	Date of borrower earliest credit line
All_enquiries	The number of times of credit enquiries
Fico_range_low	Low range of credit score
Fico_range_high	High range of credit score

Table 2: Experiment 3 Feature

Firstly, using stratified k-fold cross validation with logistic regression is a little complicated so therefore we decided to set parameter 'class_weight' to 'balanced'. We then plot the learning curve to see its learning performance. The model was trained for 12 minutes and 39 seconds.

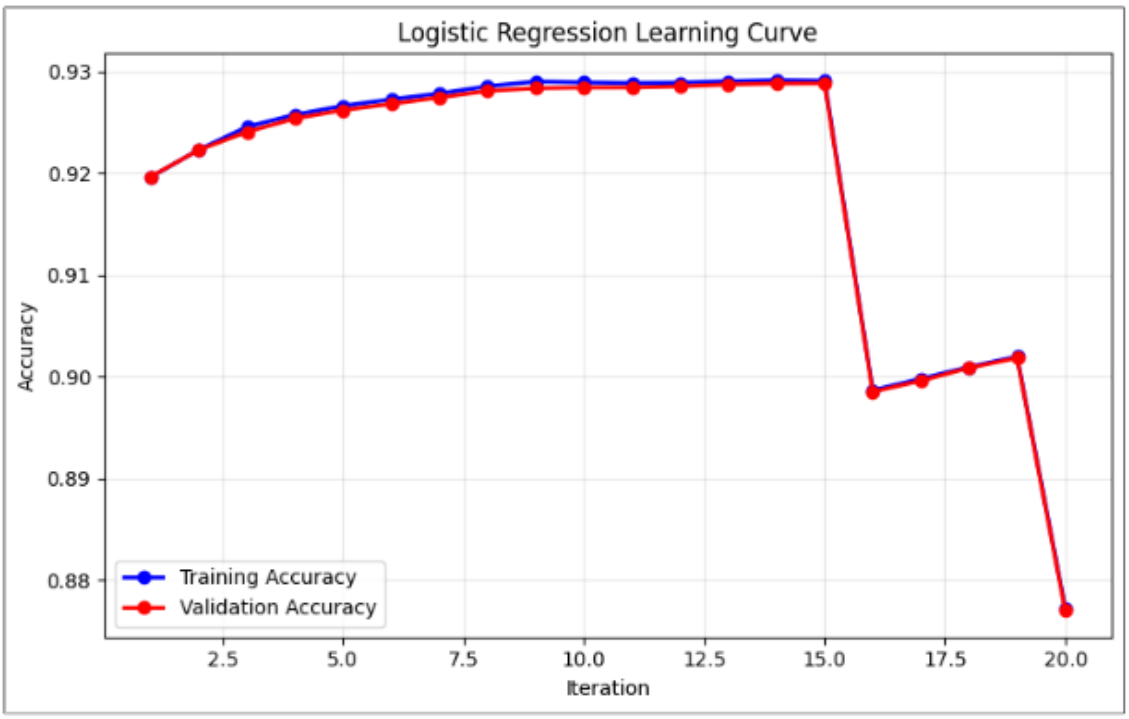


Figure 11: Experiment 3 Logistic Regression Learning Curve

From Figure 6, we can see that the training for logistic regression stop at 20 iterations with the best accuracy at 15 iterations at 92.88% accuracy during training.

Classification Report:

	precision	recall	f1-score	support
0	0.98	0.38	0.54	1196799
1	0.20	0.95	0.32	190309
accuracy			0.46	1387108
macro avg	0.59	0.66	0.43	1387108
weighted avg	0.87	0.46	0.51	1387108

Test AUC Score: 0.6632

Figure 12: Experiment 3 Logistic Regression Classification Report

However, when test on testing data, its performance is only 46% accuracy and AUC score of 0.6632.

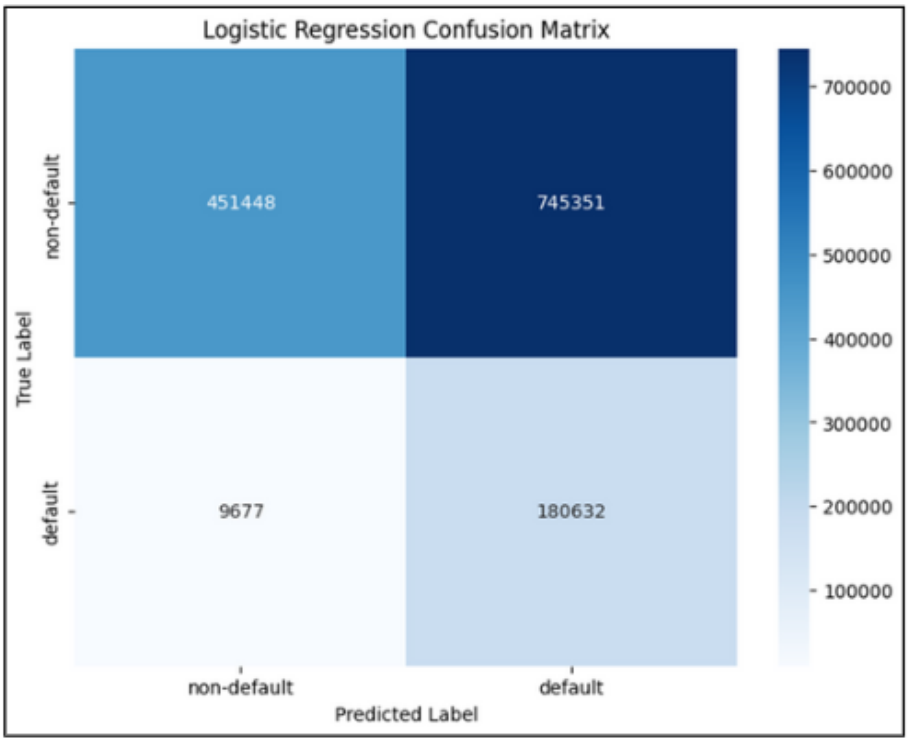


Figure 13: Experiment 3 Logistic Regression Confusion Matrix

From this matrix, we can see that although it performance on true value is quite good with minimal false positive but its false negative is too high.

Secondly, with the same LightGBM model used in previous experiments used on this feature selection. We plot its learning performance. The model was trained for 22 minutes and 50 seconds.

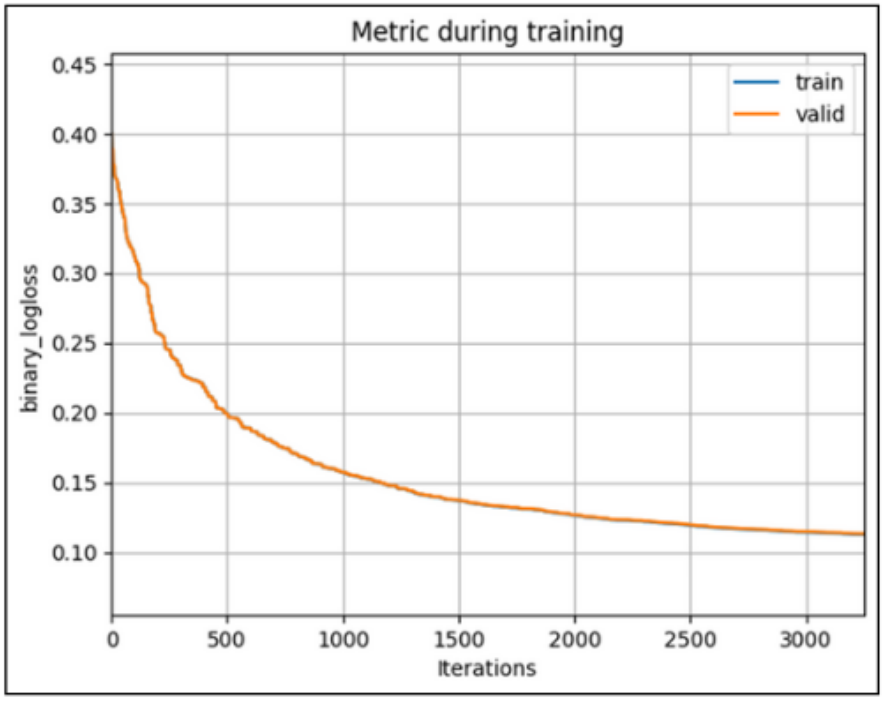


Figure 14: Experiment 3 LightGBM Learning Curve

From the graph, we can see that the model performs really well for both training and validation during training. It is at the same pace. We trained it for 5000 iterations but it stopped at just over 3000 iterations.

Accuracy: 0.9591957379011037

	precision	recall	f1-score	support
0	0.97	0.98	0.98	478677
1	0.88	0.81	0.85	76167
accuracy			0.96	554844
macro avg	0.93	0.90	0.91	554844
weighted avg	0.96	0.96	0.96	554844

Test AUC:0.8971232555988038

Figure 15: Experiment 3 LightGBM Classification Report

This is the result after evaluating on testing dataset. This model performs really well on testing data achieving an accuracy of 95.91% but the AUC score is that of 0.8971. This is due to the default data having less corrected guess than non-default.

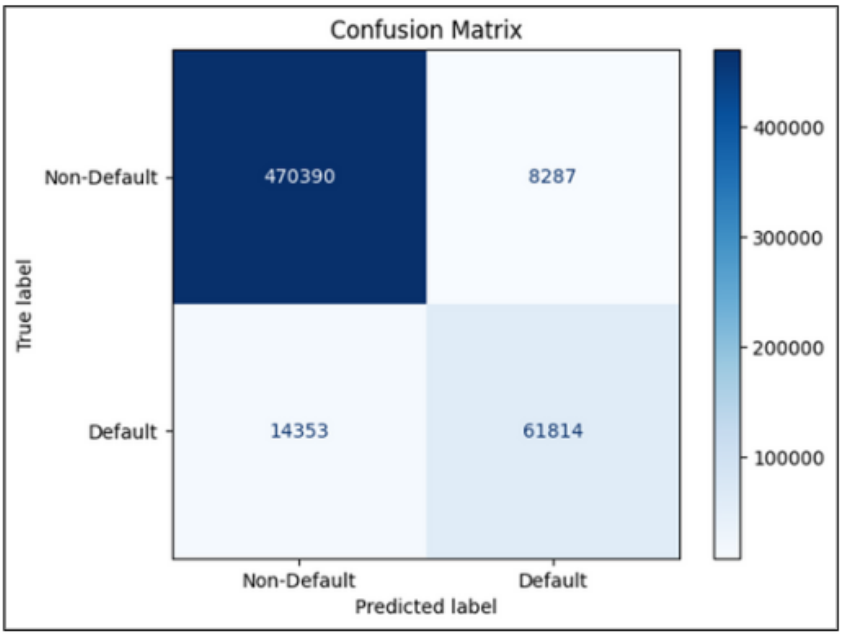


Figure 16: Experiment 3 LightGBM Confusion Matrix

This confusion matrix shows that the model has predict most of the data correctly although with some error, this model performs better than models from the experiments so far.

Last but not least, we used a neural network model to try to improve the accuracy. The model had 5 hidden layers and was used with stratified 5-fold cross validation. However, this also resulted in 5 different model for each fold that would mean that it would

take 5 times longer to process if we were to test and deploy. Therefore, We reduced it by taking out the 5-fold cross validation and use normal hold out validation instead. The model then was trained for 1 hour 13 minutes and 3 seconds.

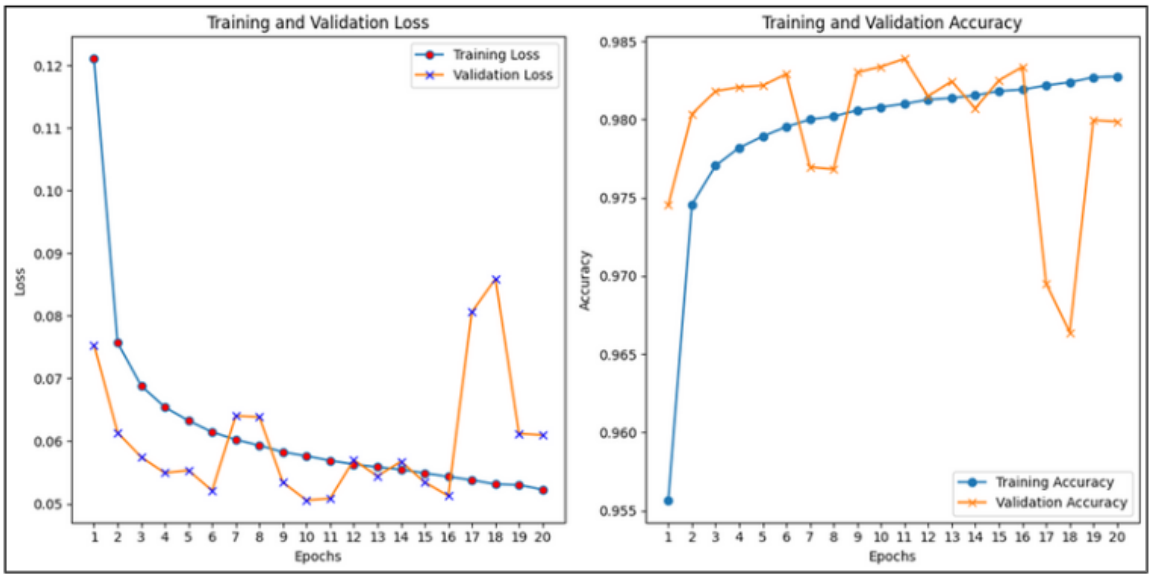


Figure 17: Experiment 3 Neural Network Training and Validation Performance

These graphs show the accuracy and loss for the model training. We focused on loss, mainly on validation loss, which from the graph on the left, the lowest that the validation loss reached is at 10 epochs. The validation loss just keeps fluctuate but never reach the same or better result as epoch 10. As for the accuracy we can see that the highest it get is around epoch 10 or 11, but as it goes on it decrease.

Accuracy: 0.9834602158444535

	precision	recall	f1-score	support
0	0.99	0.99	0.99	478677
1	0.93	0.95	0.94	76167
accuracy			0.98	554844
macro avg	0.96	0.97	0.97	554844
weighted avg	0.98	0.98	0.98	554844

AUC: 0.9957376779209361

Figure 18: Experiment 3 Neural Network Classification Report

From this classifcation report, we can see that the model perform really well on testing data reaching an accuracy of 98% and AUC score of 0.9957

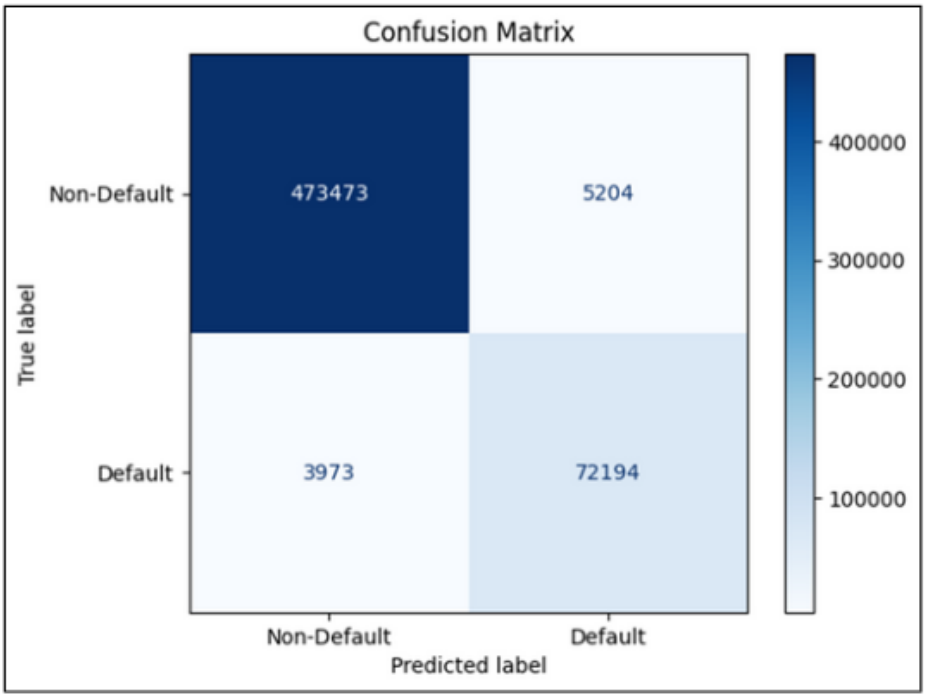


Figure 19: Experiment 3 Neural Network Confusion Matrix

From this confusion matrix, we can see that the model performs incredibly well with minimal false prediction. This model is by far the best model of all the experiments.

4.4 Model Comparison

This section evaluates the effectiveness of the proposed models, XGBoost and LightGBM, across three distinct experiments, each addressing different aspects of the loan default prediction task.

	Model	**Accuracy**	**AUC**
Literature 1	C4.5	92.64%	0.942
	Random Forest	94.55%	0.970
	Naïve Bayes	74.12%	0.813
	K Nearest Neighbor	88.85%	0.924
	Support Vector Machine (SVM)	84.38%	0.834
	Neural Network	81.75%	0.892
Literature 2	Logistic Regression	85.2%	0.797
	Naïve Bayes	77.6%	0.759
	Support Vector Machine (SVM)	85.4%	0.8
	Random Forest	86%	0.84
Literature 3	AdaBoost	71%	0.723
	Logistic Regression	67%	0.684
	Random Forest	68%	0.706

Experiment 1	LightGBM	100%	
Experiment 2	LightGBM	87%	0.613
	Logistic Regression	46%	0.663
Experiment 3	LightGBM	95.91%	0.897
	Neural Network	98.34%	0.995

Table 3: Model Comparison

From this models comparison, we can see that our highest accuracy proposed model which is neural network in experiment 3 has surpassed even the best performing model in literature review which is the RF in literature 1.

CHAPTER 5
DISCUSSION

5.1 Overview of the Discussion

This chapter delves into the insights derived from this research, emphasizing the the implications of the findings, and their significance for loan default prediction. It provides an explanation of the effectiveness of the proposed methods while highlighting valuable learnings throughout the research journey.

5.2 Discussion on Effectiveness of Proposed Models

The experimental results in Chapter 4 demonstrate varying degrees of effectiveness across models, with each approach having distinct strengths and limitations in loan default prediction. LightGBM consistently delivered strong performance, achieving 96.91% accuracy in Experiment 3, which aligns with and even surpasses several literature benchmarks. However, its susceptibility to overfitting, as observed in Experiment 1 with questionable results, shows the need for evaluation on not just training data.

The Neural Network emerged as the most effective model, achieving 98.34% accuracy and an exceptional AUC of 0.995, outperforming all other tested approaches and existing literature benchmarks. Logistic Regression, while interpretable, showed limited effectiveness (46% accuracy in Experiment 3), reinforcing that simpler linear models may lack the sophistication needed for highly imbalanced, non-linear loan default datasets. The stark contrast between its training (92.88%) and testing (46%) performance further emphasizes the risks of overfitting in less complex models.

The implementation of default prediction systems by financial institutions will be significantly impacted by these findings. Although neural networks are the most accurate, their use necessitates the right technological setup and knowledge. For organisations with limited resources or those that need more interpretable models, LightGBM offers a more approachable option that nonetheless performs exceptionally well. The particular demands of each institution should inform the choice between these strategies, taking into account variables like operational limitations, computational resources, and prediction accuracy requirements. In the end, the findings show that, when properly used and evaluated, contemporary machine learning approaches can considerably outperform conventional statistical methods in loan default prediction.

5.3 Challenges and Findings

The development and evaluation of loan default prediction models presented several notable challenges that influenced both implementation and performance. One primary difficulty was managing computational constraints, particularly for the neural network model, which required significant training time and GPU resources.

Another challenge emerged in feature selection, as the models demonstrated varying sensitivity to different feature sets. The experiments revealed that domain-specific features aligned with actual credit application processes consistently yielded better performance than generic financial indicators.

The significant performance difference between simple and complex models was one of the findings. It was evident from the logistic regression model's inability to reach a satisfactory accuracy of 46% that linear methods are insufficient for default prediction tasks. This was especially noticeable its poor generalization, since it performed radically differently throughout the testing and training periods.

Data quality and representation emerged as another significant challenge throughout the experiments. The varying results across different feature sets demonstrated how sensitive model performance is to input data characteristics. This was particularly evident in Experiment 2, where using a literature-derived feature set resulted in substantially worse performance compared to the domain-specific features used in Experiment 3.

The comparative analysis yielded practical insights for financial institutions implementing such systems. While neural networks achieved the highest accuracy, their implementation requires balancing predictive power against computational costs and interpretability needs. LightGBM offered a compelling middle ground, providing near-state-of-the-art performance with greater practicality and transparency. These findings collectively demonstrate that effective loan default prediction requires careful consideration of multiple factors beyond raw accuracy, including computational feasibility, model interpretability, and ongoing maintenance requirements.

CHAPTER 6
CONCLUSION

6.1 Summary of Findings

This study comprehensively evaluated three machine learning models—Logistic Regression (LR), LightGBM, and Neural Networks (NN)—for loan default prediction, yielding critical insights into their performance, practicality, and implementation challenges. The Neural Network demonstrated superior predictive capability, achieving 98.34% accuracy and a near-perfect AUC score of 0.995, validating deep learning's effectiveness in capturing complex, non-linear patterns in financial data. While its training process was computationally intensive (exceeding one hour), this trade-off is justifiable for high-stakes financial decisions where accuracy is paramount, and periodic model retraining is standard practice.

LightGBM emerged as the most balanced solution, delivering strong performance (95.91% accuracy) with significantly faster training times (11–22 minutes) and greater operational flexibility. Its gradient-boosted decision tree structure also offered better interpretability than neural networks, making it suitable for institutions requiring transparent decision-making. However, its occasional overfitting tendencies required careful hyperparameter tuning and validation protocols.

In contrast, Logistic Regression did poorly, with only 46% accuracy. This shows that simple linear models struggle with today's credit risk data. It couldn't handle new, unseen data well, proving that old statistical methods aren't enough for modern financial challenges.

Financial institutions must choose models based on priorities: Neural Networks for maximum accuracy where resources permit, LightGBM for efficient, high-performance deployment, and Logistic Regression as a baseline—highlighting the pitfalls of outdated methods.

This research confirms that effective loan default prediction demands more than advanced algorithms—it requires meticulous data curation and ongoing refinement. Success lies in merging cutting-edge modeling with real-world execution, ensuring theoretical power translates into practical results.

In conclusion, this study contributes to the growing field of predictive analytics in financial decision-making by providing insights into model performance, data quality, and

practical implementation. This research aims to enhance the loan approval process, supporting financial stability and economic growth in Cambodia’s lending sector.

6.2 Future Work

The findings of this study pave the way for several promising avenues of future research that can further advance loan default prediction models and their applications. A critical area for improvement lies in the access to authentic, representative datasets from Cambodian financial institutions. Collaborating with banks and financial organizations to obtain richer, more diverse datasets would allow for more accurate and context-aware models. Expanding the range of features, including borrower demographic data, economic conditions, and repayment histories, could enhance the predictive power and generalizability of the models.

Another promising direction is the development of real-time prediction systems. Current models primarily focus on batch predictions, which may not fully address the dynamic nature of loan applications and market conditions. A real-time system capable of updating predictions as new data becomes available would provide significant value to financial institutions, allowing them to make faster and more informed decisions. Implementing such systems would require the exploration of streaming data processing frameworks and lightweight, efficient models suitable for real-time applications.

While our study confirms the superior predictive performance of LightGBM and Neural Networks, their limited interpretability remains a practical challenge for financial applications. LightGBM's tree-based structure offers relatively better transparency than Neural Networks, yet both models require additional techniques like SHAP values or attention mechanisms to make their decision-making processes accessible to stakeholders and regulators. These interpretability tools could not only enhance trust in model predictions but also help identify and mitigate potential biases, addressing both operational and ethical concerns in lending decisions.

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APPENDICES

Link to our website: https://huggingface.co/spaces/PoppinGummy/loan-default-prediction